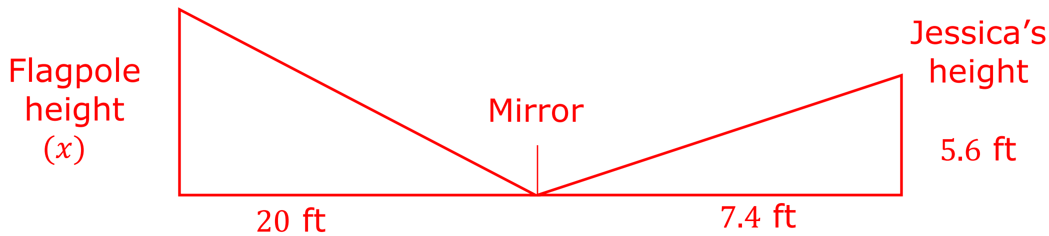
Grade 8
Unit 6
Lesson 7 Practice

Name____________________
Date____________________
Period____________________

Mrs. Williams assigned her class a project to find the height of a flagpole. The students could not measure the height, so they used a mirror to determine the height of the flagpole. Jerrica placed a mirror on the ground 20 feet from the base of the flagpole and backed up until the top of the pole was centered in the mirror. Jerrica is 5.6 feet tall and is standing 7.4 feet from the mirror.

1. Draw and label a diagram that represents this situation.

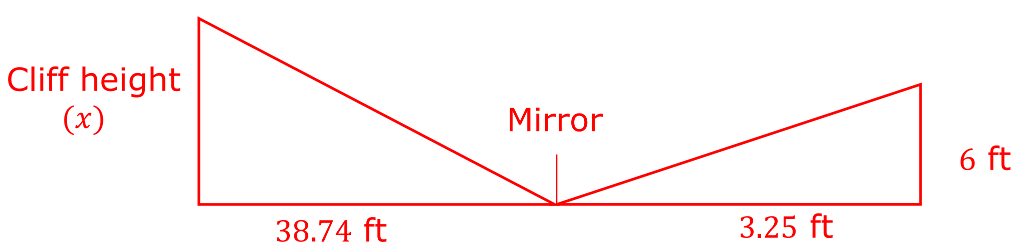


2. What is the height of the flagpole?

$\frac{x}{5.6} = \frac{20}{7.4} \rightarrow x \approx 15.135$ ft

A rock climber wants to know the height of a cliff. She places a mirror 38.74 feet from the base of the cliff, then backs up from the mirror until she can see the top of the cliff in the center of the mirror. At that point, her eye level is 6 feet above the ground and she is standing $3\frac{1}{4}$ feet from the mirror.

3. Draw and label a diagram that represents this situation.

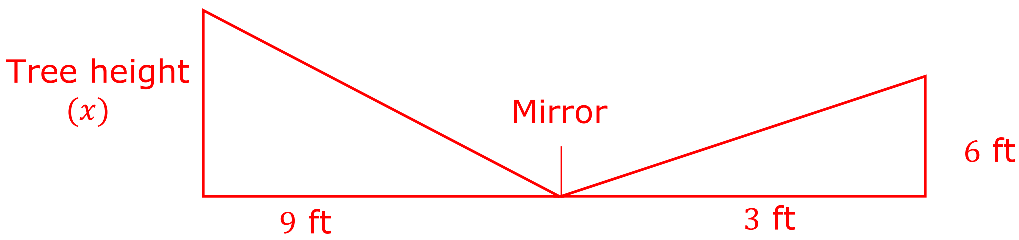


4. Find the height of the cliff to the nearest tenth of a foot.

$\frac{x}{38.74} = \frac{6}{3.25} \rightarrow x \approx 71.5$ ft

An arborist is trying to determine the height of a tree. He places a mirror 9 feet away from the tree and backs up 3 feet away from the mirror. When looking at the tree, his eye level is 6 feet from the ground.

5. Draw and label a diagram that represents this situation.

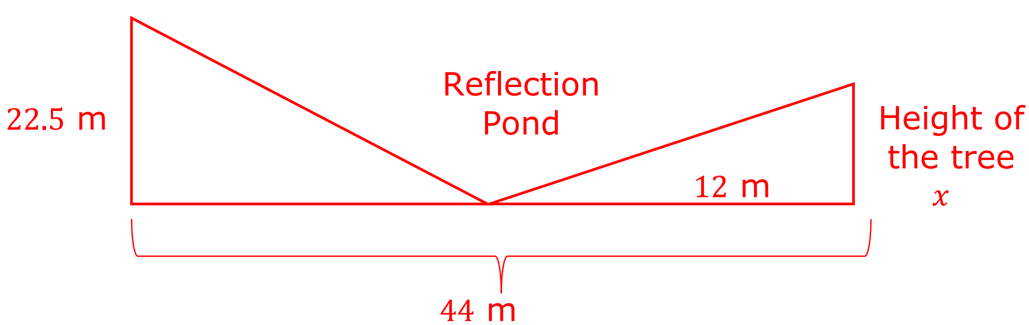


6. What is the height, in feet, of the tree?

$\frac{x}{9} = \frac{6}{3} \rightarrow x = 18$ ft

Henrietta takes her first ride in a hot air balloon. When she is 22.5 meters she looks down to notice the top of a tree is in next to a large reflection pond in a garden. She determined that the horizontal distance the hot air balloon was from the base of the tree was 44 meters and that the base of the tree was 12 meters from the center of the reflection pond.

7. Draw and label a diagram that represents this situation.

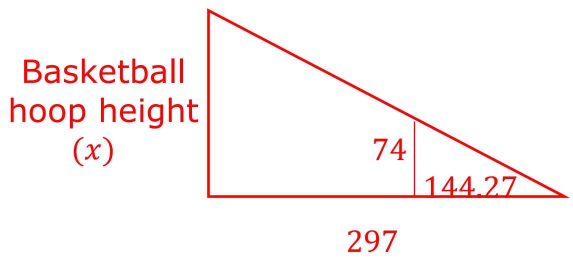


8. What is the height of the tree, in meters, to the nearest hundredth?

$\frac{22.5}{44-12} = \frac{x}{12} \rightarrow \frac{22.5}{32} = \frac{x}{12} \rightarrow \quad x \approx 8.44$ m

James is trying to find out how tall the basketball hoop is at his local park. James is 74 inches tall with a shadow length of 144.27 inches long. Using a tape measure, he was able to measure the shadow of the basketball hoop as 297 inches long.

9. Draw and label a diagram that represents this situation.



10. What is the height of the basketball hoop in ft (feet)?

$\frac{x}{297} = \frac{74}{144.27} \rightarrow x \approx 152.339$ inches ≈ 12.695 ft